MILITARY SPECIFICATION

ENAMEL, FUNGUS RESISTANT, LUSTRELESS, QUICK DRYING, STYRENATED ALKYD TYPE

1. SCOPE

1.1 Scope. This specification covers one type and one grade of quick drying olive drab lustreless enamel used as a finishing coat on ammunition and other metal surfaces. It can be used in areas covered by AIR POLLUTION REGULATIONS.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.
TT-E-516 - Enamel, Lustreless, Quick Drying Styrenated Alkyd Type.
TT-P-143 - Paint, Varnish, Lacquer and Related Materials; Packaging, Packing, and Marking of.
TT-P-442 - Pigment, Titanium Dioxide; (For Protective Coatings).
PPP-T-60 - Tape; Pressure Sensitive Adhesive, Waterproof, for Packaging and Sealing.

FSC 8010

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STANDARDS

FEDERAL

Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling and Testing.
Fed. Std. No. 595 - Colors.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Color. The color of the enamel shall be olive drab and shall match color No. 34087 of Fed. Std. No. 595 when tested as in 4.3.11.

3.2 Composition.

3.2.1 Vehicle. The vehicle shall be a styrenated phthalic alkyd resin (see 6.5.1) together with the necessary amounts of driers and volatile solvents conforming to the following requirements by volume when tested as in 4.3.3.

(a) Aromatic compounds with eight or more carbon atoms except ethylbenzene: 8% maximum.

(b) Ethylbenzene and toluene: 20% maximum.

(c) Solvents with an olefinic or cyclo-olefinic type of unsaturation: negative test (see 6.6).

(d) Ketones - negative.

(e) Total of a + b: 20% maximum.

The resin solution shall have a color value at 50 percent solids content no darker than 10 (Gardner Color Standard of 1953). Small amounts of antioxidants, wetting agents, and stabilizers may be present. The vehicle shall show a negative test for rosin and phenolic resin. The volatile solvents shall contain no benzol (benzene), chlorinated compounds or hydrolyzable chlorine derivatives.

3.2.2 Fungicide. The fungicide shall be captan (N-Trichloromethylmercapto-4-cyclohexene-1,2 dicarboximide) incorporated at a minimum of 3 percent by weight of the vehicle solids (see 6.4). When the enamel is analyzed in accordance with 4.3.6, the analysis shall conform to the requirement of Table I.

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3.2.3 Pigment. The hiding pigments shall be any combination of red or yellow iron oxide, lead chromate, titanium dioxide and carbon or lampblack. Hiding pigments shall be chemically pure and free from extenders. The titanium dioxide shall be rutile chalk resisting type conforming to type III of TT-P-442. Extender pigments shall be siliceous matter or siliceous matter and barytes and shall not exceed the amount specified in Table I. The amount of barytes shall not exceed 20 percent of the extender content by weight. Calcium sulfate or carbonate shall not be employed alone or as a component part of any pigment.

3.3 Quantatitive requirements. The enamel shall conform to requirements in Table I when tested as in 4.3.

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TABLE I - Quantitative Requirements

Characteristics	Requirements	
	Minimum	Maximum
Total solids, percent by weight of enamel	55	--
Pigment, percent by weight of enamel	33	37
Iron oxide, Fe_2O_3, percent by weight of total pigment	24 1/	--
Extender pigment, percent by weight of total pigment	--	65
Pigment volume, percent by weight of total solids	--	39
Vehicle solids, percent by weight of enamel	17	--
Phthalic anhydride, percent by weight of vehicle solids	18	26
Rosin and rosin derivatives	Negative	
Phenolic resin	Negative	
Captan, percent by weight of vehicle solids	1.3	--
Water, percent by weight of enamel	--	1.0
Coarse particles and skins, percent by weight of pigment	--	1.0
Hiding power (contrast ratio)	0.99	--
Viscosity		
Package, Krebs Stormer shearing rate - 200 r.p.m.		
Grams	125	175
Equivalent Krebs Units (K.U.)	67	77
Reduced, No. 4 Ford Cup, seconds	15	25
Fineness of grind	5	--
Specular gloss	2	8
Drying time:		
Set to touch, minutes	3	6
Dry hard, minutes	--	10
Free from after tack, minutes	--	15
Dry through, minutes	--	20
Full hardness, hours	--	72

1/ Lead chromate ($PbCrO_4$) may be substituted on an equal weight basis.

3.4 Qualitative requirements.

3.4.1 Condition in container. The enamel, tested as in 4.3.12, shall be free from grit, seeds, skins, lumps, or livering in a freshly opened full container and shall show no more pigment settling or caking than can be easily and completely reincorporated to a smooth homogeneous state.

3.4.2 Storage stability. The enamel shall show no skinning when tested as in 4.3.13 and, after aging as specified in 4.3.13 the enamel shall show no livering, curdling, tough gummy sediment, nor hard caking. It shall mix readily to a smooth homogeneous state, and any skin formed shall be continuous and easily removed.

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3.4.3 Dilution stability. When tested as in 4.3.14, the enamel shall remain stable and uniform showing no precipitation, curdling, or separation. Slight pigment settling shall be permitted.

3.4.4 Suspension properties. The enamel shall completely redisperse to a smooth homogeneous state when tested as in 4.3.15.

3.4.5 Brushing properties. The enamel, tested as in 4.3.16, shall brush satisfactorily in all respects and shall dry to a smooth, uniform film free from seeds, runs, sags, or streaks.

3.4.6 Dipping properties. When tested as in 4.3.17, the enamel shall show satisfactory dipping properties and shall present a smooth appearance, free from sagging, running, or excessive silking.

3.4.7 Spraying properties. The enamel, tested as in 4.3.18, shall spray satisfactorily in all respects and shall show no running, sagging, streaking, or blushing. The dried film shall show no dusting, mottling, or color separation, and shall present a smooth lustreless finish free from seeds.

3.4.8 Flexibility. A film of enamel tested as in 4.3.19 shall withstand bending without cracking or flaking.

3.4.9 Knife test. A film of enamel tested as in 4.3.20 shall adhere tightly and not flake, crack, or powder from the metal. The cut shall show beveled edges.

3.4.10 Adhesion. A film of enamel tested as in 4.3.21 shall show no removal of the coating by the adhesive tape beyond one-sixteenth inch on either side of the score line.

3.4.11 Water resistance. A film of enamel tested as in 4.3.22 shall show no wrinkling or blistering immediately after removal of the panel from the water. The enamel shall be no more than slightly affected when examined 2 hours after removal. After 24 hours air drying the portion of the panel which was immersed shall be almost indistinguishable with regard to hardness, color and gloss from a panel prepared at the same time but not immersed.

3.4.12 Recoating. A film of enamel tested as in 4.3.23 shall not blister, wrinkle, or show other evidence of lifting. The film shall contain no bleeding pigments which will discolor the white enamel.

3.4.13 Salt spray resistance. A film of enamel tested as in 4.3.24 and examined immediately after removal from the test shall show no rust creepage or undercutting beyond one-eighth inch from the score mark. At all other points of the panel there shall be no more than a trace of rusting (No. 9-1, method 6451 of Fed. Test Method Std. No. 141) and no more than 5 scattered blisters, none larger than 1 mm. in diameter. After removal of the enamel the surface of the steel shall show no more than a trace of rusting, pitting, or corrosion.

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3.4.14 Accelerated weathering. A film of enamel tested as in 4.3.25 shall show no more than slight chalking (No. 8, method 6411 of Fed. Test Method Std. No. 141) and a color change equivalent to a lightness difference estimate not exceeding 4 units.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Governement reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.1.1 Sampling and inspection. Sampling and inspection shall be performed in accordance with section 1000 of Fed. Test Method Std. No. 141.

4.2 Classification of tests. Testing under this specification shall be for the purpose of acceptance of individual lots. The right is reserved to make any additional tests deemed necessary to determine that the primer meets the requirements of this specification.

4.3 Test methods.

4.3.1 Test conditions. The routine and referee testing conditions shall be in accordance with section 7 of Fed. Test Method Std. No. 141 except as otherwise specified herein.

4.3.2 The following tests shall be conducted in accordance with Fed. Test Method Std. No. 141 and as hereinafter specified.

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TABLE II - Index

Tests	Test Methods: Applicable method in Fed. Std. No. 141	Test Methods: Paragraph of this specification giving further reference	Paragraph of this specification giving requirements
Color	4250	4.3.11	3.1
Benzol	5091 or 7360	--	3.2.1
Chlorinated compounds	5132	--	3.2.1
Aromatic hydrocarbon	--	4.3.3.2	3.2.1
Olefinic and cyclo-olefinic compounds	--	4.3.3.3	3.2.1
Ketones	--	4.3.3.4	3.2.1
Total solids	4041	--	Table I
Pigment solids	4022	--	Table I
Pigment analysis	4021	4.3.4	Table I
Extraction of pigment	4021	4.3.4	Table I
Iron oxide, Fe_2O_3	7271	4.3.4.1	Table I
Lead chromate, $PbCrO_4$	7131	4.3.4.2	Table I
Extender pigment, total	5271	4.3.5.1	Table I
Extender pigment analysis	7281	4.3.5.2	3.2.3
Pigment volume	4312	--	Table I
Vehicle solids	4041	--	Table I
Isolation of vehicle (super centrifuge)	4032	--	--
Phthalic anhydride	7021	--	Table I
Rosin in isolated vehicle	5031	--	Table I
Phenolic resin	5141	--	Table I
Captan	--	4.3.6	Table I
Water	4082	--	Table I
Coarse particles and skins	4092	--	Table I
Hiding power (contrast ratio)	4122	4.3.7	Table I
Viscosity	--	4.3.8	--
Package	4281	4.3.8.1	Table I
Reduced	4282	4.3.8.2	Table I
Fineness of grind	4411	--	Table I
Specular gloss, 60°	6101	4.3.9	Table I
Drying time:	4061	4.3.10	Table I
Set to touch	4061	--	Table I
Dry hard	4061	--	Table I
Free from after tack	4061	--	Table I
Dry through	4061	4.3.10.1	Table I
Full hardness	--	4.3.10.2	Table I
Condition in container	3011	4.3.12	3.4.1
Storage stability	--	4.3.13	3.4.2

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TABLE II - (Continued)

Tests	Test Methods: Applicable method in Fed. Std. No. 141	Test Methods: Paragraph of this specification giving further reference	Paragraph of this specification giving requirements
Dilution stability	--	4.3.14	3.4.3
Suspension properties	--	4.3.15	3.4.4
Brushing properties	4321, 2141	4.3.16	3.4.5
Dipping properties	4341, 2121	4.3.17	3.4.6
Spraying properties	4331, 2131	4.3.18	3.4.7
Flexibility	6221	4.3.19	3.4.8
Knife test	6304	4.3.20	3.4.9
Adhesion	--	4.3.21	3.4.10
Water resistance	6011	4.3.22	3.4.11
Recoating	--	4.3.23	3.4.12
Salt spray resistance	6061, 2011	4.3.24	3.4.13
Accelerated weathering	6152, 6122	4.3.25	3.4.14

4.3.3 Solvent analysis.

4.3.3.1 Separation of volatile portion. Pour about 15 grams of the enamel into a 50 ml. distilling flask. Add 10 ml. of tricresyl phosphate and several anti-bumping stones or Berl saddles. Fit a release valve into the mouth of the flask and attach a delivery tube to the side arm, extending into a receiver. The receiver consists of a test tube (20 x 150 mm.) with side arm for attaching to a vacuum pump. The glass delivery tube should reach 1-1/2 inches from the bottom of the tube. Immerse the receiver in a dry ice-acetone bath. Preheat a silicone oil bath to 180°C. Raise the oil bath until the oil reaches the sample level. Reduce the pressure slowly to 10 mm. of mercury. After all solvent has distilled, carefully release the vacuum using the valve that is connected to the distilling flask. Reserve the collected distillate for the aromatic solvent determination and the test for ketone, olefinic, and cyclo-olefinic compounds.

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4.3.3.2 Determination of aromatic hydrocarbons.

Apparatus: A gas chromatograph equipped with a thermal conductivity detector.

Column Preparation: Two lengths of 1/4-inch copper tubing, 6-ft. and 18-ft. long, are packed with 35% N,N-Bis(2-cyanoethyl) formamide on 60- to 80- mesh Chromosorb P.

Operating Conditions:	6-ft.	18-ft.
Detector cell temperature, °C.	300	300
Detector cell current, ma.	150	150
Injection port temperature, °C.	300	300
Helium flow at exit, cc/minute	175	110
Column temperature, °C.	125	70

4.3.3.2.1 Aromatic and oxygenated solvents - procedure A. Install the 6-ft. column and follow the operating conditions described above. Inject about 3 microliters of the isolated distillate and scan the chromatogram. The aliphatic solvents will emerge within 1 minute and the complete chromatogram should develop in about 5 minutes. From the position of the peaks observed on the chromatogram, select an internal standard that will be free of interference, such as cyclopentanol or cyclohexanol. Add 0.6 ml. of internal standard to 3 ml. of the distillate, analyze according to the above procedure. Peaks emerging after 1 minute are aromatic solvents along with any oxygenated solvents that may be present. Calculate the percent of aromatic and oxygenated solvents as follows:

$$\% \text{ aromatic and oxygenated solvents, v/v} = \frac{20^{*} \times A}{1.02^{**} \times B}$$

where, A = area of aromatic and oxygenated solvents.
B = area of internal standard.
* = percent of internal standard added.
** = is correction factor if cyclopentanol is used. If another internal standard is used, calibrate to determine the correction factor.

NOTE: If the above determination exceeds 8%, continue with the following procedure:

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4.3.3.2.2 Total aromatic content - procedure B. Place 5 ml. of the distillate in a 10-ml. glass stoppered graduate. Add 5 ml. of 85% sulfuric acid slowly while the graduate is being cooled with tap water. After the acid has been added, shake vigorously for 2 minutes then allow the layers to separate. Remove as much of the top layer as possible and wash with distilled water. Carefully pipet 3 ml. of the washed solvent into a small flask followed by 0.6 ml. of the internal standard. Mix and analyze according to procedure A. Calculate the percent of aromatics after acid treatment in the same manner as in procedure A and the percent of total aromatic solvents as follows:

$$\% \text{ total aromatic solvents, v/v} = \frac{B \times (100 - A)}{100 - B}$$

where, A = percent of aromatic and oxygenated solvents from procedure A.
B - percent of aromatic solvents after acid treatment.

NOTE: If the total aromatic content of the solvent is between 8% and 20% continue with the following procedure:

4.3.3.2.3 Toluene and ethylbenzene - procedure C. Install the 18-ft. column and follow the operating conditions described for that column. Add 0.3 ml. of high purity benzene to the 3 ml. sample used in procedure A. If the results of procedures A and B indicated the presence of oxygenated solvents, treat this sample with 85% sulfuric acid (use 3 ml. acid) as described in procedure B. Inject about 3 microliters of sample and allow the chromatograph to develop until all of the xylene isomers appear. Purge the column by raising the column temperature to 120°C. After the high boiling materials emerge reset the column temperature to 70°C. Calculate the percent of toluene and ethylbenzene as follows:

$$\% \text{ toluene, v/v} = \frac{(\text{area of toluene peak})(1.017)^{*}\ (10)^{**}}{(\text{area of benzene peak})}$$

$$\% \text{ ethylbenzene, v/v} = \frac{(\text{area of ethylbenzene peak})(1.054)^{*}\ (10)^{**}}{(\text{area of benzene peak})}$$

where, * is the correction factor for the detector response.
** is the percentage of internal standard added.

NOTE: Sensitivity of the instrument should be adjusted to keep peaks from running off the scale. Appropriate corrections must be made for changes in sensitivity when computing the peak areas.

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4.3.3.3 Test for olefinic or cyclo-olefinic compounds. Take 2 test tubes and place 2 drops of the distillate in each. Dissolve the first sample in 1 ml. of carbon tetrachloride and add 1 drop of 1 percent bromine in carbon tetrachloride. Shake and allow to set for 5 minutes. A positive test is indicated by the complete absence of yellow color when observed against a white background. Dissolve the second sample in 1 ml. of acetone and add 1 drop of 1 percent permanganate solution (1 gram of potassium permanaganate crystals in 95 mls. of acetone and 5 mls. of water). Shake and allow to set for 2 minutes. A positive test is indicated by the decolorization of the purple solution. The solvent is considered to fail the test for olefinic and cyclo-olefinic compounds if either of the above tests is positive (see 3.2.1.2 and 6.6).

4.3.3.4 Test for ketones.

4.3.3.4.1 Reagent. Two grams of 2,4-dinitrophenylhydrazine + 4 mls. of concentrated sulfuric acid + 30 mls. methanol (add slowly) + 10 mls. water.

4.3.3.4.2 Procedure. Pipette 1 ml. of reagent into a 20 x 170 mm. test tube. Add 10 drops of distillate (see 4.3.3.1) and shake for 30 seconds. A yellow precipitate or cloud in the reagent layer indicates the presence of ketones. Run a blank using one milliliter of reagent and 10 drops of mineral spirits.

4.3.4 Analysis of pigment. Extract the pigment as in method 4021 of Fed. Test Method Std. No. 141 using extraction mixture C.

4.3.4.1 Iron oxide (Fe_2O_3) content. Determine the iron oxide (Fe_2O_3) content on the extracted pigment in accordance with method 7141 of Fed. Test Method Std. No. 141.

4.3.4.2 Lead chromate ($PbCrO_4$) content. Determine the lead chromate ($PbCrO_4$) content on the extracted pigment in accordance with 4.2.1, method 7131 of Fed. Test Method Std. No. 141.

4.3.5 Extender pigment.

4.3.5.1 Total extender pigment content. Determine matter insoluble in acid in the extracted pigment by method 5271 of Fed. Test Method Std. No. 141.

4.3.5.2 Extender pigment analysis. Determine barium sulfate, siliceous material, and calcium (sulfate or carbonate) by the applicable portions of method 7281 of Fed. Test Method Std. No. 141.

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4.3.6 Captan analysis. From a tared dropping bottle add about 1 gram of the isolated vehicle dropwise to 50-ml. of heptane which is being agitated with a magnetic stirrer. Re-weigh the bottle and calculate the sample weight. Filter the heptane solution quantitatively, collecting the filtrate and washings in a 100-ml. beaker. Evaporate the filtrate to dryness on a steam bath. Redissolve the residue in chloroform and transfer it quantitatively to a 25-ml. volumetric flask and dilute to mark. Transfer a 1 ml. aliquot of this chloroform solution to a test tube, 175 x 22 mm., which contains 0.3 gram of resorcinol. Evaporate the chloroform using a water bath, 45° - 55°C., and a stream of air. Transfer the test tube to an oil bath, 135°C. ± 1°C., for 15 minutes, rotating the tube periodically. Remove the test tube from the oil bath and immediately dilute the contents with ethyl alcohol containing 10% by volume of 0.1 N aqueous sulfuric acid. Transfer the sample quantitatively to a 25-ml. volumetric flask and dilute to the mark with acidic ethanol. Let the color develop for one hour. Filter the solution through paper of fine porosity into another 25-ml. flask using a vented cork and watch glass cover over the funnel to minimize evaporation. Discard the first 10 or 15 ml. of filtrate. Read the absorbance of the solution at 425 nm against a blank carried through the above procedure. Determine the concentration of captan from a working curve obtained by carrying known amounts of captan (0.2 to 0.4 milligrams) through the above procedure. Report the percent captan on a nonvolatile vehicle basis.

4.3.7 Hiding power (contrast ratio). Determine the contrast ratio in accordance with method 4122 of Fed. Test Method Std. No. 141. Use a film applicator that will deposit a 3 inch wide film with a dry film thickness of 0.0010 inch maximum and air dry for 72 hours. Determine the reflectance and verify the film thickness in the area in which the reflectance was measured. Calculate the contrast ratio and check for compliance with Table I.

4.3.8 Viscosity.

4.3.8.1 Package. Proceed as in method 4281 of Fed. Test Method Std. No. 141 except that the method of mixing shall be to agitate the can for 3 minutes on a paint shaker[1/].

[1/] An apparatus of this type, powered by a 1/4-hp motor, operates at a rate of 1350 shakes per minute, and is manufactured by Red Devil Tools, Irvington, N. J.

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4.3.8.2 Reduced viscosity. Reduce 3 parts by volume of enamel with one part by volume of thinner conforming to Table III and determine viscosity as in method 4282 of Fed. Test Method Std. No. 141. Check for compliance with Table I.

Table III - Thinner

Ingredient	Percent by Weight
VMP Naphtha (8% max. aromatic)	65
n-Butyl alcohol	20
Toluene	15

4.3.9 60° specular gloss. Draw down the package material using a 0.002 inch (0.004 inch gap clearance) film applicator. Measure the gloss as specified in method 6101 of Fed. Test Method Std. No. 141.

4.3.10 Drying time. Determine drying time in accordance with method 4061 of Fed. Test Method Std. No. 141 under referee conditions, except that the enamel shall be drawn down with a 0.002 inch (0.004 inch gap clearance) film applicator.

4.3.10.1 Dry through. Determine dry through in accordance with method 4061 except the enamel shall be drawn down with a 0.002 inch (0.004 inch gap clearance) film applicator on a 4 by 12 inch steel panel that has been phosphoric acid etched as in method 2011, procedure B of Fed. Test Method Std. No. 141.

4.3.10.2 Full hardness. Determine full hardness on a panel prepared as in 4.3.10.1. The film shall be considered to have reached full hardness when it is very difficult to remove with a knife blade.

4.3.11 Color. In accordance with method 4250 of Fed. Test Method Std. No. 141, compare the color with the film of enamel on the white carrara glass panel prepared for the hiding power test and observe for compliance with 3.1.

4.3.12 Condition in container. Determine package condition in accordance with method 3011 of Fed. Test Method Std. No. 141 and observe for compliance with 3.4.1.

4.3.13 Storage stability. Determine 48 hour skinning in accordance with method 3021 of Fed. Test Method Std. No. 141 and observe for compliance with 3.4.2. Reseal and age for 7 days at 60°C. and observe for compliance with 3.4.2.

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4.3.14 Dilution stability. Reduce one volume of packaged enamel with one volume of thinner conforming to Table III slowly adding the thinner to the enamel with constant stirring. Allow to stand 24 hours, thoroughly remix, and using a 0.0005 inch (0.0010-inch-gap clearance) film applicator, draw down a film of the mixture on clear plate glass. Before and after the film is dry, examine by transmitted light for compliance with 3.4.3.

4.3.15 Suspension properties. Reduce the enamel as in 4.3.8.2. Place 6 ounces of the reduced enamel in an 8-ounce glass jar. Allow the stoppered jar to remain undisturbed for 24 hours and then place the unopened jar on a paint shaker as in 4.3.12 and agitate the contents for 20 seconds. Re-examine the material for any evidence of nonhomogeneity or undispersed pigment. Observe for compliance with 3.4.4.

4.3.16 Brushing properties. Apply the enamel to a 4- by 12-inch steel panel, using a 1-1/2 inch brush, and observe for brushing properties in accordance with method 4321 of Fed. Test Method Std. No. 141 for compliance with 3.4.5.

4.3.17 Dipping properties. Reduce three parts by volume of enamel with one part by volume of thinner conforming to Table III. Test the dipping properties of the thinned enamel in accordance with method 4341 of Fed. Test Method Std. No. 141 and observe for compliance with 3.4.6.

4.3.18 Spraying properties. Reduce the enamel as in 4.3.8.2. Spray on a steel panel to give a dry film thickness of 0.0009 to 0.0011 inch and observe for spraying properties in accordance with method 4331 of Fed. Test Method Std. No. 141 and observe for compliance with 3.4.7. For referee test use automatic application per method 2131 of Fed. Test Method Std. No. 141.

4.3.19 Flexibility. Determine flexibility in accordance with method 6221 of Fed. Test Method Std. No. 141. Draw down a 2-inch wide film of enamel with suitable film applicator that will give a dry film thickness of 0.0009 to 0.0011 inch on a 31 gage (0.0107 inch) cold rolled lustre-finish steel panel prepared in accordance with method 2011 of Fed. Test Method Std. No. 141 using the petroleum naphtha ethylene glycol monoethylether mixture. Allow the test panel to air dry 1/2 hour and then bake for 24 hours at 105° ± 2°C. (221° ± 4°F.). Condition the panel for 1/2 hour at 25° ± 1°C. Bend over a 1/4 inch mandrel and examine for compliance with 3.4.8.

4.3.20 Knife test. Perform the knife test in accordance with method 6304 of Fed. Test Method Std. No. 141 using a flat portion of the baked panel from the flexibility test. Observe for compliance with 3.4.9.

4.3.21 Adhesion, tape test.

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4.3.21.1 Panel preparation. Using a 0.0020-inch (0.0040-inch gap clearance) film applicator, draw down a 2-inch wide film of the enamel on a steel panel, solvent cleaned and phosphoric acid etched as in procedure B, method 2011 of Fed. Test Method Std. No. 141.

4.3.21.2 Procedure. Air dry the specimen for 1 hour under referee conditions and then score a line through to the metal across the width of the film using a sharp pointed knife. The film shall then be taped perpendicular to and across the score line with waterproof, pressure-sensitive adhesive tape (3/4 inch wide) conforming to PPP-T-60, type IV. The tape shall be pressed in firm contact with the film and shall extend for approximately 1 inch on each side of the score line. All air bubbles shall be rolled out by firm pressure of the thumb. Allow approximately 10 seconds for the test area to return to room temperature. Grasp a free end of the tape, at a rapid speed, strip it from the specimen by pulling the tape back upon itself at an angle of 180°. Observe the specimen for compliance with 3.4.10.

4.3.22 Water resistance. Prepare two panels as in 4.3.21 and air dry for 72 hours. Coat all exposed, uncoated metal surfaces with wax or other suitable coating and immerse one of the panels for 18 hours in distilled water at 23° ± 1°C. (73.4° ± 2°F.) in accordance with method 6011 of Fed. Test Method Std. No. 141. At the end of the test period remove the panel from the water and inspect for compliance with 3.4.11.

4.3.23 Recoating. Prepare two panels as in 4.3.21 and air dry for 24 and 72 hours, respectively. At the end of its drying period immerse each panel to a depth of 2-1/2 inches in a white enamel conforming to specification TT-E-516, composition L which has been reduced with one part by volume of thinner conforming to Table III to four parts by volume of package material. At the end of 5 seconds remove the panel, dry in a vertical position, and examine for compliance with 3.4.12.

4.3.24 Salt spray resistance. Solvent clean three 4- by 12-inch steel panels in accordance with 2011 of Fed. Test Method Std. No. 141 and apply a phosphate coating conforming to TT-C-490, type I. Reduce the enamel as in 4.3.8.2 and spray the test panels to a uniform dry film thickness of 0.0009 to 0.0011 inch. Air dry for 72 hours, score, and expose to 5 percent salt spray for 48 hours as in method 6061 of Fed. Test Method Std. No. 141. Upon removal, wash the panels gently in running water not warmer than 100°F. until free from any visible salt deposits and examine immediately for compliance with 3.4.13. Strip the enamel film from the panels by means of lacquer thinner and inspect the steel for rust, pitting, or corrosion.

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4.3.25 Accelerated weathering. Prepare a film of the enamel as in 4.3.19 and air dry for 72 hours. Measure the directional reflectance and expose the panel to accelerated weathering for 168 hours in accordance with method 6152 of Fed. Test Method Std. No. 141 using a twin arc apparatus. Examine the exposed panel for chalking by rubbing with a piece of velvet or cheesecloth wrapped around the finger. Using moderate pressure, draw the cloth across the width of the panel in two different directions. Measure the directional reflectance (method 6121) on an unrubbed area of the exposed panel and determine the amount of color change, expressed as lightness index difference (ΔL), using method 6122 of Fed. Test Method Std. No. 141. Check for compliance with 3.4.14.

4.3.26 Packaging, packing, and marking. The enamel shall be inspected for compliance with the packaging, packing, and marking requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. The enamel shall be packaged, packed, and marked in accordance with TT-P-143. The level of packaging shall be A, B or C, and the level of packing shall be A, B, or C, as specified (see 6.2). The enamel shall be furnished in 1-quart or 1-gallon multiple friction top containers, in 5-gallon lug cover steel pails or in 55-gallon steel drums, as specified (see 6.2).

6. NOTES

6.1 Intended use. The enamel covered by this specification is intended for use as a finish coat on phosphated or primed metal surfaces and on ammunition components. IT IS NOT INTENDED for use where gasoline resistance is a prime requisite such as automotive equipment.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Size of container required (see section 5).

(c) Level of packaging and level of packing (see section 5).

6.3 The enamel covered by this specification should be purchased by volume, the unit being one U.S. liquid gallon of 231 cubic inches at 68°F. (20°C.).

6.4 Due to partial breakdown of the fungicide in preparation of the enamel, it is necessary to incorporate a minimum of 3 percent captan (non-volatile vehicle basis) to meet the 1.3 percent requirement by analysis.

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6.5 The olive drab enamel is contemplated to be comparable in performance to the following approximate composition by weight:

280	lbs. Synthetic yellow orange iron oxide
13.5	lbs. Carbon black
72.5	lbs. Rutile titanium dioxide
2.5	lbs. Synthetic medium red iron oxide
190	lbs. Fibrous magnesium silicate
445	lbs. Acicular talc
18	lbs. Captan
430	lbs. Styrenated alkyd resin (R-T at 50 percent solids in xylol)
550	lbs. Xylol
3	lbs. Diethylamine

Grind 18 - 24 hours in a porcelain ball mill using a 2 - 1 ratio of porcelain balls by weight and reduce as follows:

755	lbs. Styrenated alkyd resin (R-T at 50 percent solids in xylol)
155	lbs. Xylol
3	lbs. Cobalt naphthenate
3	lbs. Antiskinning agent

At low viscosities the use of small amounts of suspension agents will improve settling properties.

6.5.1 The styrenated alkyd resins used in these formulations are copolymers of styrene and phthalic alkyd resins modified with drying vegetable oils. It has generally been found that the finished enamel must be finely ground in order to ensure adequate salt spray resistance and good ballistic performance when applied on ammuniton.

6.5.2 Enamels of this type show some tendency to be thixotropic at higher viscosities. The use of a material such as diethylamine to the extent of 0.5 to 1.0 percent on the resin solids basis has been found to reduce this condition materially and improve the package stability.

6.6 The test for olefinic and cyclo-olefinic compounds will not be positive for solvents containing less than 1 percent of these compounds.

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